

Vivek Gangurde

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ML Engineer

SECTOR 18, GURGAON, HARYANA

ML ENGINEER

Experienced ML Engineer with 3 years of expertise in ML Algorithms(Supervise and Unsupervised), Deep learning and NLP. Knowledge of Python Data Analysis and Machine Learning Libraries, TensorFlow, Keras, Pytorch and OpenCV for computer vision. Worked on tool like - Jupyter Notebook, Visual studio, Flask, yolov8. Have expertise in preparing data, developing and deploying highly scalable ML, DL and NLP models.

WORK EXPERIENCE

Truefan (aug 2024 - Present)

Audio-Driven 3D Lip-Sync Model Enhancement

- Integrated 3D Morphable Model (3DMM) for improved facial geometry in lip-sync animations.
- Added HuBERT-based audio feature extraction for better phoneme-to-motion alignment.
- Enhanced training loss functions with LPIPS (Perceptual Loss using VGG), MSE, and LMD Loss, improving lip synchronization accuracy.
- Fine-tuned lip region generation to increase lip-sync quality and realism.

Shivacha (Dec 2023 - July2024)

TTS and Voice cloning

- Implementation: Successfully implemented TTS and voice cloning using Open Source TTS model.
- Fine-Tuning: Improved voice cloning and speech generation quality for Hindi, achieving a mean average precision of 91 %.
- Enhancements: Increased character limit for text input from 250 to 1000.
- Bulk Audio Generation: Successfully implemented bulk audio generation for large-scale tasks.
- Developed a real-time TTS and lip-sync system, integrating TTS and Retrieval-Based Voice Conversion (RVC) models

Hudl (Oct 2021- Nov 2023)

Development of a web app for Kabaddi analysis. Player & Raider Tracking and speed estimation.

- Developed custom deep learning model for player & raider detection and tracking.
- Integrated video processing for match analysis.speed calculation algorithms for player attacks.
- Utilized Streamlit for interactive dashboards. Included video links for post-match analysis

Formation and Backfield Detection for American Football

- Developed a custom computer vision model utilizing YOLOv8 architecture to accurately detect formations and backfield players in American football footage.
- Collaborated with domain experts to gather and annotate the football formation dataset, ensuring high-quality and accurate labels for training.
- Achieved an impressive mean average precision (mAP) of 92% through meticulous model optimization and hyperparameter tuning.
- Reduced around 50% manual tagging time for analysts, thereby enhancing workflow efficiency and allowing coaches to access accurate and actionable insights promptly.

SKILLS

Programming Languages: Python NumPy, Pandas, PyTorch, TensorFlow, Plotly, Streamlit, GAN , Matlab

Computer Vision: OpenCV, CNN, Vision Transformers, YOLO, TensorFlow, Keras. 3D model

ML & NLP : Algorithms(Supervise and Unsupervised),Scikit-learn, Seaborn, Matplotlib, RNN, LSTM, Transformer

Tools: Git, MySQL, Flask, AWS

Languages: English, Hindi, Marathi.

EDUCATION & CERTIFICATIONS

Savitribai Phule Pune University

Bachelor of Engineering Mechanical Engineering

Pune Vidyarthi Griha's COET

Completed in 2018

PERSONAL DETAILS

DOB: 14/05/1994

Languages: English, Hindi, Marathi

Status: Unmarried

EXTRACURRICULAR ACTIVITIES

Team Lead.

Community Sports Events: Organized and participated in community sports events such as soccer tournaments, Cricket tournaments, basketball, fostering community engagement and promoting healthy lifestyles.

